List of Symbols

Abbreviations

A Algorithm

C Corollary

D Definition

E Example

EPf End of proof

F Fact

L Lemma

LS Left side

RS Right side

N Notation

OM Order matters

~~OM~~ Order does not matter

P Proposition

Pf Proof

Q Question

RA Repetition allowed

~~RA~~ Repetition is not allowed

S Specification

SPf Start of proof

T Theorem

W Witness

Special Sets

$\mathbb{N}$ The set of natural numbers starting at 0

$\mathbb{Z}$ The set of whole numbers

$\mathbb{Q}$ The set of rational numbers

$\mathbb{R}$ The set of real numbers

$\mathbb{C}$ The set of complex numbers

Logic

$=$ is equal

$:=$ is defined to be equal to; is, by definition, equal to (what stands on the left of this symbol is being defined and what stands on the right is the definition)

$\neq$ is not equal to

$\neg$ not

$\wedge$ and

$\vee$ or

$\Longrightarrow$ implies

$\Longleftrightarrow$ if and only if, implies and is implied by

$:\Longleftrightarrow$ by definition, if and only if (what stands on the left of this symbol is being defined and what stands on the right is the definition)

$\exists$ for some, there exists some

$\nexists$ for no, there does not exist any

$\forall$ for every

$\in$ is in, is contained in, belongs to

$\notin$ is not in, is not contained in, does not belong to

$\subseteq$ is included in, is a subset of

$\nsubseteq$ is not included in, is not a subset of

$\cup$	Union
$\biguplus$	Disjoint union
$\cap$	Intersection
$\setminus$	Without (Relative complementation)
$\sqcap$,$\times$	Cartesian Product
$\sqcup$	Coproduct
$\because$	Since, because
$\therefore$	Hence, therefore